

Using a demo shopping website such as SauceDemo (Swag Labs), perform Selenium WebDriver automation for login, product selection, cart, and checkout operations. For alert handling, mouse actions, drag-and-drop, and dynamic-element handling, use a suitable Selenium demo website that supports these interactions.

Selenium WebDriver Exercises

Test Case	Shopping Scenario	Selenium Concept	Expected Result
TC01	Open the online shopping website	driver.get()	Shopping website opens successfully
TC02	Customer clicks Delete/Remove Product and confirmation popup appears	Alert – accept()	Product deletion is confirmed
TC03	Customer clicks Delete/Remove Product but chooses Cancel	Alert – dismiss()	Product remains in the cart
TC04	Customer enters a name/coupon/customer information in a prompt popup	Prompt – send_keys()	Entered information is submitted successfully
TC05	Customer moves the mouse over the Products/Category menu	Mouse Hover	Product categories/submenu are displayed
TC06	Customer double-clicks a product	Double Click	Product details page opens
TC07	Customer drags a product/item into a shopping cart area	Drag & Drop	Product is moved to the cart
TC08	Customer searches for a product and waits for the product results to load	Explicit Wait	Product is displayed successfully
TC09	Customer completes checkout and waits until the Place Order button becomes clickable	Clickable Wait	Order is submitted successfully
TC10	Customer completes the purchase and waits for the order confirmation popup	Alert Wait	Confirmation alert is handled successfully

PROGRAM:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.action_chains import ActionChains
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

driver = webdriver.Chrome()
driver.maximize_window()

wait = WebDriverWait(driver, 10)

driver.get("https://www.saucedemo.com/")

print("TC01 - Shopping website opened successfully")

wait.until(
    EC.visibility_of_element_located((By.ID, "user-name"))
).send_keys("standard_user")

driver.find_element(By.ID, "password").send_keys("secret_sauce")
driver.find_element(By.ID, "login-button").click()

wait.until(
    EC.visibility_of_element_located((By.CLASS_NAME, "inventory_list"))
)

print("TC01 - Login successful")
```

```
driver.get("https://www.selenium.dev/selenium/web/alerts.html")
```

```
driver.find_element(By.ID, "alert").click()
```

```
alert = wait.until(  
    EC.alert_is_present()  
)
```

```
print("\nTC02 - Alert:")
```

```
print(alert.text)
```

```
alert.accept()
```

```
print("TC02 - Product deletion confirmed")
```

```
driver.get("https://www.selenium.dev/selenium/web/alerts.html")
```

```
driver.find_element(By.ID, "confirm").click()
```

```
alert = wait.until(  
    EC.alert_is_present()  
)
```

```
print("\nTC03 - Confirmation Alert:")
```

```
print(alert.text)
```

```
alert.dismiss()
```

```
print("TC03 - Product remains in the cart")
```

```
driver.get("https://www.selenium.dev/selenium/web/alerts.html")
```

```
driver.find_element(By.ID, "prompt").click()
```

```
alert = wait.until(  
    EC.alert_is_present()  
)
```

```
print("\nTC04 - Prompt Alert:")
```

```
print(alert.text)
```

```
alert.send_keys("Aditya")
```

```
alert.accept()
```

```
print("TC04 - Customer information submitted successfully")
```

```
driver.get("https://www.selenium.dev/selenium/web/mouse_interaction.html")
```

```
hover_element = wait.until(  
    EC.visibility_of_element_located((By.ID, "hover"))  
)
```

```
ActionChains(driver).move_to_element(hover_element).perform()
```

```
move_status = wait.until(  
    EC.visibility_of_element_located((By.ID, "move-status"))  
)
```

```
print("\nTC05 - Mouse Hover performed successfully")  
print("Result:", move_status.text)
```

```
driver.get("https://www.selenium.dev/selenium/web/mouse_interaction.html")
```

```
double_click_element = wait.until(  
    EC.visibility_of_element_located((By.ID, "clickable"))  
)
```

```
ActionChains(driver).double_click(double_click_element).perform()
```

```
click_status = wait.until(  
    EC.visibility_of_element_located((By.ID, "click-status"))  
)
```

```
print("\nTC06 - Double Click performed successfully")  
print("Result:", click_status.text)
```

```
driver.get("https://www.selenium.dev/selenium/web/mouse_interaction.html")
```

```
source = wait.until(  
    EC.visibility_of_element_located((By.ID, "draggable"))
```

```
)
```

```
target = wait.until(
```

```
    EC.visibility_of_element_located((By.ID, "draggable"))
```

```
)
```

```
ActionChains(driver).drag_and_drop(source, target).perform()
```

```
drop_status = wait.until(
```

```
    EC.visibility_of_element_located((By.ID, "drop-status"))
```

```
)
```

```
print("\nTC07 - Drag and Drop performed successfully")
```

```
print("Result:", drop_status.text)
```

```
driver.get("https://www.automationexercise.com/products")
```

```
search_box = wait.until(
```

```
    EC.visibility_of_element_located((By.ID, "search_product"))
```

```
)
```

```
search_box.clear()
```

```
search_box.send_keys("Top")
```

```
search_button = wait.until(
```

```
    EC.element_to_be_clickable((By.ID, "submit_search"))
```

```
)
```

```
search_button.click()
```

```
searched_products = wait.until(
    EC.visibility_of_element_located(
        (
            By.XPATH,
            "//h2[contains(translate(normalize-space(.), 'abcdefghijklmnopqrstuvwxyz',
'ABCDEFGHIJKLMNOPQRSTUVWXYZ'), 'SEARCHED PRODUCTS')]"
        )
    )
)
```

```
products = wait.until(
    EC.presence_of_all_elements_located(
        (
            By.XPATH,
            "//div[contains(@class,'productinfo')]"
        )
    )
)
```

```
print("\nTC08 - Product search completed successfully")
print("Result:", searched_products.text)
print("Number of products found:", len(products))
```

```
driver.get("https://www.saucedemo.com/")
```

```
wait.until(
```

```
    EC.visibility_of_element_located((By.ID, "user-name"))
).send_keys("standard_user")

driver.find_element(By.ID, "password").send_keys("secret_sauce")
driver.find_element(By.ID, "login-button").click()

wait.until(
    EC.visibility_of_element_located((By.CLASS_NAME, "inventory_list"))
)

driver.find_element(
    By.ID, "add-to-cart-sauce-labs-backpack"
).click()

driver.find_element(
    By.CLASS_NAME, "shopping_cart_link"
).click()

wait.until(
    EC.visibility_of_element_located((By.CLASS_NAME, "cart_item"))
)

driver.find_element(By.ID, "checkout").click()

wait.until(
    EC.visibility_of_element_located((By.ID, "first-name"))
).send_keys("Aditya")

driver.find_element(By.ID, "last-name").send_keys("S")
```

```
driver.find_element(By.ID, "postal-code").send_keys("600001")
```

```
driver.find_element(By.ID, "continue").click()
```

```
finish_button = wait.until(  
    EC.element_to_be_clickable((By.ID, "finish"))  
)
```

```
print("\nTC09 - Place Order button is clickable")
```

```
finish_button.click()
```

```
confirmation = wait.until(  
    EC.visibility_of_element_located(  
        (By.CLASS_NAME, "complete-header")  
    )  
)
```

```
print("TC09 - Order submitted successfully")
```

```
print("Result:", confirmation.text)
```

```
driver.get("https://www.saucedemo.com/")
```

```
wait.until(  
    EC.visibility_of_element_located((By.ID, "user-name"))  
)  
.send_keys("standard_user")
```

```
driver.find_element(By.ID, "password").send_keys("secret_sauce")
```

```
driver.find_element(By.ID, "login-button").click()
```

```
wait.until(  
    EC.visibility_of_element_located((By.CLASS_NAME, "inventory_list"))  
)
```

```
driver.find_element(  
    By.ID, "add-to-cart-sauce-labs-backpack"  
)click()
```

```
driver.find_element(  
    By.CLASS_NAME, "shopping_cart_link"  
)click()
```

```
wait.until(  
    EC.visibility_of_element_located((By.CLASS_NAME, "cart_item"))  
)
```

```
driver.find_element(By.ID, "checkout").click()
```

```
wait.until(  
    EC.visibility_of_element_located((By.ID, "first-name"))  
)send_keys("JAYARAJ")
```

```
driver.find_element(By.ID, "last-name").send_keys("B")  
driver.find_element(By.ID, "postal-code").send_keys("600123")
```

```
driver.find_element(By.ID, "continue").click()
```

```
finish_button = wait.until(
    EC.element_to_be_clickable((By.ID, "finish"))
)
```

```
finish_button.click()
```

```
wait.until(
    EC.visibility_of_element_located(
        (By.CLASS_NAME, "complete-header")
    )
)
```

```
driver.execute_script(
    "setTimeout(function(){alert('Order confirmation successful')}, 500)"
)
```

```
alert = wait.until(
    EC.alert_is_present()
)
```

```
print("\nTC10 - Order Confirmation Alert:")
print(alert.text)
```

```
alert.accept()
```

```
print("TC10 - Confirmation alert handled successfully")
```

```
time.sleep(2)
```

```
driver.quit()
```

```
print("\nAll test cases completed successfully")
```

OUTPUT:

```

Problems Output Debug Console Terminal Ports
(.venv) PS C:\Users\ladini\OneDrive\Desktop\Selenium> c:\ cd "c:\Users\ladini\OneDrive\Desktop\Selenium" & "c:\Users\ladini\OneDrive\Desktop\Selenium\venv\Scripts\python.exe" "c:\Users\ladini\vscode\extensions\ms-python.debugpy-2023.6.8-win32-x64\bundled\libs\debugpy\launcher" 57458 -f "c:\Users\ladini\OneDrive\Desktop\Selenium\demo_site.py"
(.venv) PS C:\Users\ladini\OneDrive\Desktop\Selenium> c:\ cd "c:\Users\ladini\OneDrive\Desktop\Selenium" & "c:\Users\ladini\OneDrive\Desktop\Selenium\venv\Scripts\python.exe" "c:\Users\ladini\vscode\extensions\ms-python.debugpy-2023.6.8-win32-x64\bundled\libs\debugpy\launcher" 48993 -f "c:\Users\ladini\OneDrive\Desktop\Selenium\demo_site.py"
TC01 - Shopping website opened successfully
TC01 - Login successful
TC02 - Alert:
       Cheems
TC02 - Product deletion confirmed
TC03 - Confirmation Alert:
Are you sure?
TC03 - Product remains in the cart
TC04 - Prompt Alert:
Enter something
TC04 - Customer information submitted successfully
TC05 - Mouse Hover performed successfully
Result: hovered
TC06 - Double Click performed successfully
Result: double-clicked
TC07 - Drag and Drop performed successfully
Result: dropped
TC08 - Product search completed successfully
Result: SEARCHED PRODUCT'S
Number of products found: 14
TC09 - Place Order button is clickable
TC09 - Order submitted successfully
Result: Thank you for your order!
TC10 - Order Confirmation Alert:
Order confirmation successful
TC10 - Confirmation alert handled successfully
All test cases completed successfully
(.venv) PS C:\Users\ladini\OneDrive\Desktop\Selenium>

```

